

TCGA Multimodal Cancer Survival Analysis Project

Project Overview

This project implements a comprehensive multimodal deep learning approach for cancer survival prediction using data from The Cancer Genome Atlas (TCGA). The system integrates clinical, genomic, and text data to predict patient survival outcomes with state-of-the-art machine learning techniques.

1. Data Acquisition

1.1 Clinical Data from cBioPortal

- **Source:** cBioPortal for Cancer Genomics
- **Data Retrieved:**
 - 9,524 patient samples with 82 clinical features
 - Key features: Age, Sex, Race, TNM staging, Pathological stages, Treatment history
 - Survival outcomes: Overall Survival (OS_MONTHS, OS_STATUS)
- **Format:** CSV file with comprehensive clinical annotations

1.2 Gene Expression Data from GDC Repository

- **Source:** Genomic Data Commons (GDC) Repository
- **Data Type:** RNA-Seq expression profiles
- **Initial Size:** ~60,000 genes across 8,459 samples
- **Processing:**
 - Downloaded FPKM-normalized expression values
 - Filtered to include only **protein-binding genes**
 - Final gene set: ~5,000 protein-binding genes
- **Rationale:** Protein-binding genes are functionally important for cancer progression and treatment response

1.3 Mutation Data from GDC Repository

- **Source:** GDC Repository mutation annotation files
- **Data Type:** Somatic mutation calls
- **Processing Steps:**
 1. Parsed MAF (Mutation Annotation Format) files
 2. Created binary mutation matrix (1 = mutated, 0 = not mutated)
 3. Included all genes with at least one mutation across cohort
- **Final Format:** Binary matrix of 8,459 samples $\times$ ~1,000 mutated genes

1.4 Pathological Text Embeddings

- **Source:** Clinical pathology reports from TCGA
 - **Embedding Model:** Bio-ClinicalBERT (Alsentzer et al., 2019)
 - Pre-trained on clinical notes from MIMIC-III
 - Fine-tuned on clinical/biomedical text
 - **Processing:**
 - Extracted pathology report text
 - Generated 768-dimensional embeddings using ClinicalBERT
 - Created embedding matrix: 9,523 samples $\times$ 768 features
 - **Output:** `bio_clinicalbert_embeddings.csv`
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2. Data Integration and Preprocessing

2.1 Sample Alignment

- **Challenge:** Unequal sample sizes across modalities
 - Clinical: 9,824 samples
 - Expression: 8,459 samples
 - Mutation: 8,459 samples
 - Text embeddings: 9,523 samples
- **Solution:**
 - Identified 8,459 common samples across all modalities
 - Implemented missing modality handling for partial samples

2.2 Feature Engineering

Clinical Features:

- Age squared (non-linear age effects)
- TNM composite score (weighted combination)
- Binary indicators (metastasis, node involvement)
- One-hot encoding for categorical variables

Expression Features:

- Top 5,000 most variable protein-binding genes
- Log2 transformation and standardization
- Batch effect correction

Mutation Features:

- Binary encoding (mutation present/absent)

- Feature selection based on mutation frequency

2.3 Handling Imbalanced Data

- **Problem:** Low event rate (~20% deaths)
 - **Solutions Implemented:**
 1. **Focal Loss:** Adaptive weighting for hard examples
 2. **Weighted Sampling:** 2.5x weight for death events
 3. **Stratified Splitting:** Maintained event ratio across splits
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3. Model Architectures

3.1 MissingAwareMultimodalAutoencoder

Architecture:

- Modality-specific encoders with different architectures
- Attention-based fusion mechanism
- Handles missing modalities gracefully

Key Features:

- Clinical encoder: 3-layer MLP (input $\rightarrow$ 512 $\rightarrow$ 256 $\rightarrow$ latent_dim)
- Expression encoder: Deep network (input $\rightarrow$ 2048 $\rightarrow$ 1024 $\rightarrow$ 512 $\rightarrow$ latent_dim)
- Mutation encoder: 2-layer network (input $\rightarrow$ 1024 $\rightarrow$ 512 $\rightarrow$ latent_dim)
- Text encoder: 2-layer network (768 $\rightarrow$ 512 $\rightarrow$ 256 $\rightarrow$ latent_dim)
- Multi-head attention (8 heads) for cross-modal learning

Loss Function: Cox Proportional Hazards or Focal Survival Loss

3.2 RobustTransformerSurvival

Architecture:

- Transformer-based multimodal fusion
- Learnable missing modality tokens
- Global aggregation token

Key Components:

- Modality embeddings: Linear projection to d_model=512
- Transformer layers: 4 layers with cross-attention
- Position encoding for modalities

- Auxiliary event prediction task

Innovation: Uses cross-attention between global token and modality tokens

3.3 AdaptiveEnsembleSurvival

Architecture:

- Ensemble of Autoencoder and Transformer models
- Adaptive weighting based on available modalities
- Meta-learner for final prediction

Key Features:

- Dynamic weight prediction network
 - Modality availability-aware ensemble
 - Final prediction: $0.7 \times \text{weighted_ensemble} + 0.3 \times \text{meta_learner}$
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4. Training Strategy

4.1 Loss Functions

1. **Cox Proportional Hazards Loss:** Standard survival loss
2. **Focal Survival Loss:**
 - $\alpha = 0.3, \gamma = 2.0$
 - Focuses on hard-to-classify samples
3. **Ranking-Aware DeepHit Loss:**
 - Combines ranking and likelihood losses
 - Higher weight (2x) for event pairs

4.2 Optimization

- **Optimizer:** AdamW with weight decay $1e-4$
- **Learning Rate:** $1e-4$ (reduced to $1e-5$ for BERT layers)
- **Scheduler:** ReduceLROnPlateau (patience=10)
- **Gradient Clipping:** 1.0
- **Early Stopping:** Patience=20 on validation C-index

4.3 Data Augmentation

- Modality dropout (10% probability)
- Gaussian noise for continuous features
- Random masking for missing modality robustness

5. Evaluation Metrics

5.1 Primary Metrics

- **Concordance Index (C-index):** Main performance metric
- **Time-dependent AUC:** At 6, 12, 24, 36, 60 months
- **Integrated Brier Score:** Calibration measure

5.2 Risk Stratification

- **Log-rank Test:** Between risk groups (Low/Medium/High)
 - **Kaplan-Meier Curves:** Visual survival analysis
 - **Hazard Ratios:** Between risk groups
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6. Key Innovations

6.1 Missing Modality Handling

- Learnable missing tokens for Transformer
- Modality-specific gates in Autoencoder
- Availability-aware attention masking

6.2 Imbalance Strategies

- Focal loss adaptation for survival
- Event-aware weighted sampling

6.3 Multimodal Integration

- Cross-modal attention mechanisms
 - Adaptive ensemble based on data availability
 - Hierarchical feature fusion
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7. Implementation Details

7.1 Technology Stack

- **Deep Learning:** PyTorch, PyTorch Lightning

- **Survival Analysis:** Lifelines
- **Data Processing:** Pandas, NumPy, Scikit-learn
- **Visualization:** Matplotlib, Seaborn, Plotly
- **Text Processing:** Transformers (HuggingFace)

7.2 Computational Requirements

- **GPU:** NVIDIA GPU with 16GB+ VRAM recommended
 - **Training Time:** ~2-4 hours on single GPU
 - **Memory:** 128GB RAM for data preprocessing
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8. Future Work Models

8.1 Model

Model	C-Index	Events C-Index
Clinical Only	0.82	0.80
Multimodal Autoencoder	In progress	In progress
Transformer	In progress	In progress
Adaptive Ensemble	In progress	In progress

8.2 Feature Importance

1. **Clinical:** TNM stage, Age, Treatment history
2. **Expression:** Cell cycle genes, Immune markers
3. **Mutations:** TP53, KRAS, EGFR
4. **Text:** Treatment response, Histological grade

8.3 Clinical Applicability

- Risk stratification for treatment planning
 - Identification of high-risk patients
 - Biomarker discovery for targeted therapy
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